The examination is being carried out on the **following application documents**

Description, Pages

1-77 as published

Claims, Numbers

1-15 filed with entry into the regional phase before the EPO

Drawings, Sheets

1/14-14/14 as published

Reference is made to the following documents; the numbering will be adhered to in the rest of the procedure.

D1 MENG-YAO LI ET AL: "Real-time monitoring of antibody glycosylation site occupancy by in situ Raman spectroscopy during bioreactor CHO cell cultures", BIOTECHNOLOGY PROGRESS, vol. 34, no. 2, 21 January 2018 (2018-01-21), pages 486-493, XP072295789

D2 DANIEL A ZAVALA-ORTIZ ET AL: "Interest of locally weighted regression to overcome nonlinear effects during in situ NIR monitoring of CHO cell culture parameters and antibody glycosylation", BIOTECHNOLOGY PROGRESS, vol. 36, article no. e2924, 14 October 2019 (2019-10-14), pages 1-10, XP072291181

D3 BRANDON BERRY ET AL: "Cross-scale predictive modeling of CHO cell culture growth and metabolites using Raman spectroscopy and multivariate analysis", BIOTECHNOLOGY PROGRESS, vol. 31, no. 2, 29 December 2014 (2014-12-29), pages 566-577, XP072293416

D4 RUTH C ROWLAND-JONES ET AL: "Spectroscopy integration to miniature bioreactors and large scale production bioreactors-Increasing current capabilities and model transfer", BIOTECHNOLOGY PROGRESS, vol. 37, article no. e3074,19 September 2020 (2020-09-19), pages 1-14, XP072300068

D5 WO 2016/004322 A2 (BIOGEN MA INC [US]) 7 January 2016 (2016-01-07)

D6 Wang Yan ET AL: "Monitoring Glycosylation Profile and Protein Titer in Cell Culture Samples Using ZipChip CE-MS", Journal of Analytical & Bioanalytical Techniques- 3rd International Conference and Exhibition on Analytical & Bioanalytical Techniques November 22-24, 2012 Hyderabad International Convention Centre, India, vol. 08, no. 02, 1 January 2017 (2017-01-01), pages 1-8, XP093302453

1 The amendments filed on 20.10.2023 with entry into the regional phase before the EPO are acceptable in the view of the requirements of Art. 123(2) EPC; they are based on published claims 1-43.

2 Independent claims 1, 7, 10 and 14 are not supported by the description as required by Article 84 EPC, as their scope is broader than justified by the description. The claimed subject-matter encompasses *any* glycated molecule, whereas support within the meaning of Article 84 EPC and disclosure within the meaning of Article 83 EPC has been given only for glycated proteins. Furthermore, the claims are only supported in as far as the PAT tool is Raman spectroscopy. The disclosed findings are only based on this type of spectroscopy for allowing to develop models that can predict the protein structure solely on a measured spectrum. Thus, said claims are not supported and disclosed over their whole breadth.

In addition, claims 1 and 7 are only supported in as far as the culturing of cells in a culture medium for producing the glycated protein is specified.

3 For the subject-matter of claims 1-15 which is considered to be sufficiently supported (see above objection), novelty in the sense of Art. 54 EPC and the presence of an inventive step in the sense of Article 56 EPC is recognized.

3.1 The document D1 discloses (see abstract):

A novel methodology for real-time monitoring of antibody glycosylation site occupancy using Raman spectroscopy is proposed. It was first observed in CHO cell batch culture that when low nutrient concentrations were reached, a decrease in monoclonal antibody (mAb) glycosylation was induced, which made it essential to rapidly detect this loss of product quality. By combining in situ Raman spectroscopy with chemometric tools, efficient prediction models were then developed for both glycosylated and nonglycosylated mAbs. By comparing variable importance in projection profiles of the prediction models, it was confirmed that Raman spectroscopy is a powerful method to distinguish extremely similar molecules, despite the high complexity of the culture medium. Finally, the Raman prediction models were used to monitor batch and feed-

harvest cultures in situ. For the first time, it was demonstrated that the concentrations of glycosylated and nonglycosylated mAbs could be successfully and simultaneously estimated in real time with high accuracy, including their sudden variations due to medium exchanges. Raman spectroscopy can thus be considered as a promising PAT tool for feedback process control dedicated to on-line optimization of mAb quality.

The partial least squares (PLS) regression was adopted for developing the prediction models (see page 488, left-hand column, last paragraph- right-hand column, 2nd paragraph).

The cultures were grown in bioreactors having a volume of 2 L (see page 487, right-hand column, paragraph 3).

Fermentation runs were performed in a first set of bioreactors (2 L volume), wherein spectra were measured, and from which spectra the regression model was developed (see page 488, right-hand column, last paragraph - page 490, right-hand column, second-last paragraph; figures 1 and 2).

A further series of runs was performed in a second set of bioreactors (2 L volume) to validate the model, in particular for runs performed in batch and feed-harvest modes, which were intended to be representative of industrial processes. Spectra were measured and the level of glycation (total (T) mAB versus non-glycosylated (NG) mAB) was determined based on the model previously developed (see page 490, right-hand column, last paragraph - page 492, left-hand column, first paragraph; figures 3 and 4).

It is noted that in present claim 1, the actual volumes corresponding to the indicated first and second threshold volumes of the first and second bioreactors are not specified, nor is the relation of the first to the second threshold specified. Hence, according to claim 1, the first and second bioreactors may have the same or different values, and the first bioreactor may be larger, smaller or of the same size as the second bioreactor. The method of D1, using first and second bioreactors of the same size, therefore comprises this feature specified in claim 1.

The method according to independent claim 1 or independent claim 7 therefore differs from the method of D1 in that it measures and determines the levels of glycan structures (opposed to only a level of glycation without any structure information). The subject-matter of claims 1 and 7 is therefore novel. The method of claim 7 additionally differs from the method of D1 in that the level of a glycan structure is measured in a bioreactor having a volume of equal to or above 1,000 liters.

3.2 The document D2 is a publication by the same research group as D1, and it similarly comprises (see abstract of D2): the measuring of spectra, the development of regression models, and the validation of the models in cell cultures for predicting in real time the level of glycosylation of produced antibodies. The difference versus D1 is that in the method of D2 near infrared spectroscopy (NIR) is used instead of Raman spectroscopy. D2 (see abstract) further discloses that, whereas monitoring models are mainly constructed using PLS regressions (PLSR), evidences presented in this study indicate nonlinearity relationship between in situ captured spectra and compound concentrations, compromising the PLSR performances. A novel and simple approach was proposed to fit non-linearity using the locally weighted regression (LWR). The LWR models were found to be more appropriate for handling information contained in spectra so that real time monitoring of cultures were accurately performed. Moreover, for the first time, the LWR calibration models allowed mAb glycosylation to be monitored, in a real time manner, by using in situ NIR spectroscopy.

The cultures were grown in bioreactors having a volume of 2 L, with 1.5 L working volume (see paragraph 2.1).

Fermentation runs were performed in a first set of bioreactors (2 L volume), wherein spectra were measured, and from which spectra the regression model was developed (see paragraph 3.1 and figure 1). A further series of runs was performed in a second set of bioreactors (2 L volume) to validate the model, wherein spectra were measured and the level of glycation (total versus non-glycosylated mAB) was determined based on the model previously developed (see paragraph 3.3; figure 2).

The subject-matter of independent claims 1 and 7 presents the same differences against D2 as those indicate above in comparison to D1.

3.3 Concerning claims 10 and 14, the documents D1 and D2 (as summarized above) also disclose a system for producing a glycosylated molecule comprising:

Means for culturing a glycosylated molecule-producing cell line (2L bioreactor for culturing CHO cells expressing mAB, see e.g. page 487 in D1); means for measuring a level of glycation, wherein the means generates spectral data (Raman spectrometer, see e.g. BioViserion®, specified on page 488 in D1); means for generating one or more regression models based on the spectral data (see e.g. PLS-toolbox ver. 8.1, specified on page 488 in D1); and means for measuring a level of glycation in a glycosylated molecule-producing cell line (same as the Raman spectrometer).

Or, worded differently:

A bioreactor comprising a glycosylated molecule-producing cell line (2L bioreactor for culturing CHO cells expressing mAB, see e.g. page 487 in D1); a process analytical technology (PAT) tool that measures glycation and generates spectral data (Raman spectrometer, see e.g. BioViserion®, specified on page 488 in D1); and a processor that correlates levels of glycation with the spectral data using one or more regression models (see e.g. PLS-toolbox ver. 8.1, specified on page 488 in D1).

However, D1 or D2 do not disclose means for measuring a level of one or more glycan structures.

The subject-matter of claims 10 and 14 is therefore novel in the view of D1 or D2.

3.4 The subject-matter of independent claims 1, 7, 10 and 14, as far as it is sufficiently supported (see above objection), is considered to involve an inventive step for the following reasons.

The technical effect of the above-identified difference, i.e. measuring and determining levels of glycan structures, is that the method provides a more detailed readout. For example, the levels of different glycoforms of a glycosylated protein can be determined.

The problem to be solved can thus be formulated as to provide a more detailed quality analysis of the glycated products.

The solution provided in the claims is considered to involve an inventive step for the following reasons.

D1 (page 488, right-hand column, last paragraph) does mention that in the production of glycosylated mABs, variations of glycosylation occur at two major levels, namely (i) the presence or absence of oligosaccharide structures at the glycosylation sites (macroheterogeneity), or (ii) the presence of different oligosaccharide structures at one specific glycosylation site (microheterogeneity). In their study, the authors found that the major variations in mAb glycosylation during the CHO cell culture in batch mode came from glycan site occupancy (>30%) compared with oligosaccharide structures (<15%). The authors therefore focused their study on macroheterogeneity of the mAb glycosylation.

D1 (page 490, right-hand column, paragraph 2) further identifies the peaks at 1126, 1129, and 1139 cm^{-1} of the Raman spectrum that can be assigned to N-acetyl glucosamine, fucose, and mannose, respectively, which are three major glycan residues representative of mAb N-glycosylation.

However, D1 does not suggest that accurate measurement of the levels of glycan structures may be obtained online during a fermentation run using the same methodology based on Raman spectroscopy and regression modelling using the PLS regression.

Similarly, neither D2 nor any other prior art method has discussed or suggested the possibility of determining glycan structures in this way.

Conversely, D6 discloses (see abstract) a method of monitoring the glycosylation profile (structure) in cell culture samples using capillary electrophoresis (CE) coupled to mass spectrometry (MS). The method allows to measure the presence and relative level of glycoforms of produced antibodies, such as G0F or G0F-GlcNAc, G1F forms (see fig. 6). However, the method of D6 does not rely on spectral data alone and it does not use regression modelling to determine the structures from spectral data.

3.5 Claims 2-6, 8, 9, 11-13 and 15 are dependent on claims 1, 7, 10 and 14, respectively, and as such also meet the requirements of the EPC with respect to novelty and inventive step.

3.6 The documents D3-D5 are cited as additional background prior art, especially for the subject-matter relating to the use of bioreactors having different sizes, including manufacturing scale.

3.6.1 In D3 (see abstract), multi-component, multi-scale Raman spectroscopy modeling results from a monoclonal antibody producing CHO cell culture process including data from two development scales (3 L, 200 L) and a clinical manufacturing scale environment (2,000 L) are presented. An array of single-scale and combination-scale modeling iterations were generated to evaluate technology capabilities and model scalability. The conclusions in D3 (see page 576) are that Raman spectroscopy appears to be well suited for realtime predictions of many important nutrient, metabolic waste, and cell growth parameters within a mammalian cell culture setting. While glucose, lactate, and osmolality models exhibited the best performance with all scenarios of combination-scale calibration models, other models such as glutamate and ammonium models are less reliable, and the models of viable cell density (VCD), and total cell density (TCD) have a clear scale dependency. It is

indicated that end users should develop a strong understanding of the performance characteristics of the PLS models they generate to identify where weaknesses may exist in cross-scale models.

3.6.2 D4, which refers back and cites the publication of D3, as well that of D2, is a further development for the application of models across different scales. It discloses (abstract; page 2, right-hand column, paragraph 2) an integrated solution to a miniature bioreactor system (ambr®15; working volume of 15 ml) that not only enables fully automated setup for Raman spectral acquisition but also facilitates model transfer across scales by using a spectral probe head with the same optical path, as well as an integrated setup to ambr®15 (static) through to on-line monitoring at 50 L scale (non-static) that can be applied to commercial scale vessels (see also the results presented in paragraph 3.7, figs. 8 and 9).

3.6.3 D5 is a patent document by some of the same authors as D3; it discloses (claim 1):

A method of assessing a bioreactor culture, the method comprising:

(i) obtaining a Raman spectrum of a manufacturing-scale bioreactor culture; and

(ii) determining a culture parameter of the manufacturing-scale bioreactor culture using a model that relates the Raman spectrum to the culture parameter, wherein the model is developed based on one or more test bioreactor cultures of a smaller volume than the manufacturing-scale bioreactor culture.

The volume of the manufacturing-scale bioreactor culture may be in a range of 1000 L to 4000 L (claim 2).

The volume of the test bioreactor culture may be in a range of 1 L to 400 L (claim 3).

The model may be a partial least squares model (claim 8).

Although D5 does not disclose the direct monitoring of glycation level by spectroscopy, it mentions (see page 25) the issue of highly variable nutrient (e.g. glucose) levels leading to erratic protein glycosylation, sialylation as well as other key product quality attributes.

4 The application does not meet the requirements of Article 84 EPC, because claims 1 and 7 are not clear.

Claim 1 indicates that the method comprises a "plurality of runs" within a bioreactor. However the claim does not specify that in the bioreactor, cells producing a glycosylated molecule are cultured. Hence a "run" (which must be a fermentation run or the like) is not clearly defined in the claim.

The same objection applies to claim 7 ("trial runs").